

Wearable Alert Device: BOM

University of Oklahoma
School of Electrical and Computer Engineering
Spring 2026 Semester
Senior Capstone Project

Team Members:

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Written By:
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Item	Item Description	Quantity	Cost \$	Use	Part Number
1	Arduino Nano ESP32	1	\$19.30	Main Micro controller for project	Mouser - ABX00083
2	Lithium Ion Battery Pack - 3.7V 4400mAh	1	\$19.95	Main Power supply for project	Adafruit - 354
3	Micro-Lipo Charger for LiPoly Batt with USB Type C Jack	1	\$5.95	Rechargeable module for Battery Pack	Adafruit - 4410
4	Gravity: GNSS GPS BeiDou Receiver Module - I2C&UART	1	\$17.90	GPS Module for Project	Digikey - 1738-TEL0157-ND
5	FERMION MEMS MICROPHONE MODULE(Analog)	1	\$2.90	Microphone Module for GS Detection	Digikey - SEN0487
6	320E11RED	1	\$2.09	Button for panic device	Digikey - 101391
7	Testing LED (Red)	1	\$0.55	LED used to tell user if device is working	Digikey - SLR-56VC3F
8	Decoupling Capacitor (0.1 Microfarad)	1	\$0.89	Used to remove noise from battery pack power	Digikey - C322C104K3G5TA
9	Current Limiting Resistor(1 KOhm)	1	\$0.10	Prevents blow out of LED	Digikey - CFF14JT1K00
10	Wires(Jumper/Cut) (Spool)	1	\$15.89	Came with boards but used to connect device	Half came with the Micro breadboards/ other half from link
11	Proto Boards	1	\$0.28	Used to more permanently seat device	ELEGOO 32 Pcs Double Sided PCB Board Prototype Kit
12	Electrical Tape(Roll)	1	\$0.97	Used to hold device together before case	Hyper Tough Electrical Tape Single Roll 60 Feet Long Black
		Total:	\$86.77		